

SYLLABUS
INTRODUCTION TO TIME SERIES ANALYSIS
FALL 2026

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

Course Title: Introduction to Time Series Analysis (STAT 429)

Instructor: Sabyasachi Chatterjee

Institution: University of Illinois Urbana-Champaign

Lectures: TBA

Office Hours: TBA

Teaching Assistants: TBA

Lab Sections: TBA

TA Office Hours: TBA

About the course:

Time series (and more generally sequential data) refers to datasets where there is a temporal order for the observations. This course aims to teach you how to analyze such data. The primary objective of time series analysis is to develop mathematical models that provide plausible descriptions for observed time series data. We shall study several time series models in this class including (a) Multiple linear regression models (covariates depending on time), (b) Nonlinear regression models (covariates depending on time), (c) Regularized High-dimensional linear regression models (covariates depending on time), (d) Variance models and spectral analysis, (e) Lagged regressions and ARIMA models, (f) Recurrent Neural Networks.

Prerequisites:

Undergraduate probability and statistics is required at the level of 411 is required. Regression at the level of 425 is strongly recommended.

Programming Language:

You are free to use any language (e.g., R, Python, Julia, Matlab etc.) for working on your homework. We will be using Python code in class and the lab sections.

Text:

Date: August 2026.

There is no required textbook for the class. Here are some materials that you can use as general references:

- Lecture materials from Aditya Guntuboyina's Fall 2025 offering are available at <https://stat153.berkeley.edu/spring-2025/>. This semester's course will closely follow that version.
- Prof. Ryan Tibshirani's lecture materials for STAT 153 in Fall 2024 are available at <https://stat153.berkeley.edu/fall-2024/>.
- *Time Series Analysis and its Applications* by Shumway and Stoffer. It has good materials on ARIMA modeling and Spectral Analysis.

I will provide materials for each lecture (including slides or typed lecture notes, and code) which will be posted after the lecture.

Discussion Platform: TBA

Homework assignments:

There will be five homework assignments during the semester. The schedule for the homework assignments will be announced in class.

Exams:

There will be two exams: Midterm and Final. The Midterm will be held during the ninth week of the course, tentatively on October 21, 2026. The date and time of the Final Exam are TBA.

Assessment:

Your final score for the class will be calculated as the maximum of:

$$50\% \text{ Homework} + 20\% \text{ Midterm} + 30\% \text{ Final Exam}$$

and

$$50\% \text{ Homework} + 50\% \text{ Final Exam}.$$

Each homework assignment is worth an equal amount (we will not have a policy of dropping the lowest homework).

Grade Complaints:

If you have a complaint against an assigned homework or exam grade and want to talk to me about it, first send me a written request through email explaining your case clearly.

Academic Integrity:

Collaboration in small groups on homework problems is encouraged, but your write-up must be entirely your own, and you may not read or copy another student's solutions. You may consult books, online resources, and generative AI tools (e.g., ChatGPT or other LLMs) to support your learning; if you do, you must (i) acknowledge all such sources/tools in your write-up and (ii) ensure you fully understand and can independently reproduce and explain every step of your solution. Copying text, code, or solutions—verbatim or paraphrased— from any source (including AI tools) without clear attribution, or submitting work you do not understand, is cheating. Students found to be cheating risk failing the course and being referred to the appropriate University office.

How to cite AI briefly (example): “Consulted ChatGPT for hints on Problem 4; adapted approach after verifying steps.”

Students with disabilities:

If you need accommodations for any physical, psychological, or learning disability, please get in touch with me so that we can make the necessary arrangements.